



Cohort canonical evaluation — 40 agents

idea-01-lateral-attribution · 534 held-out segments · V0
yaw 0.01627 rad/s, CTE 254.26 m · generated
2026-06-01 10:12

- Executive summary

AGENTS (OK / TOTAL)

39 / 40

EVAL POOL

534 segs

V0 YAW RMSE

0.01627

V0 CTE RMSE

254.26 m

WALL TIME

16.88s

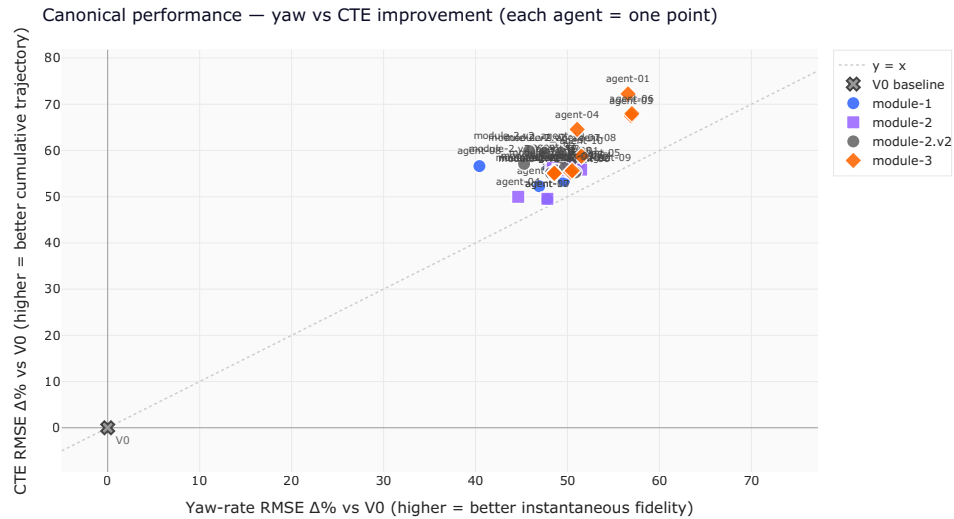
Best yaw: m3-agent-06 **+57.0%** · Best CTE: m3-agent-01

+72.2%

Winning both KPIs $\geq +30\%$ (39 agents): m3-agent-06, m3-agent-03, m3-agent-01, m3-agent-10, m2-agent-05, m3-agent-04, m1-agent-01, module-2.v2_agent-09, m3-agent-09, m3-agent-08, m2-agent-08, m1-agent-09, module-2.v2_agent-05, m1-agent-07, module-2.v2_agent-08, m2-agent-10, module-2.v2_agent-04, m1-agent-02, m1-agent-05, m2-agent-01, module-2.v2_agent-06, m3-agent-07, m3-agent-02, module-2.v2_agent-02, m2-agent-03, m1-agent-03, m2-agent-02, m2-agent-09, m2-agent-06, module-2.v2_agent-01, m1-agent-04, m2-agent-07, m1-agent-10, module-2.v2_agent-07, m1-agent-06, module-2.v2_agent-10, module-2.v2_agent-03, m2-agent-04, m1-agent-08

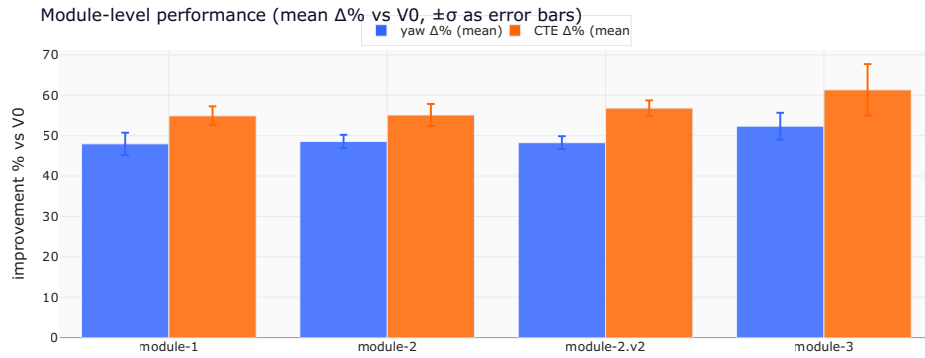
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is $y = x$: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

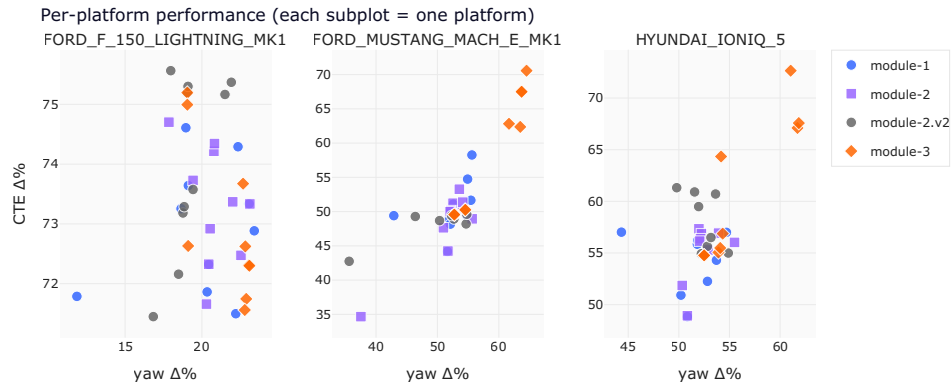
Performance grouped by which module shipped the agent.
Mean $\Delta\%$ over only the agents that reconstructed successfully;
failures shown separately.



family	n ok / total	yaw $\Delta\%$ (mean $\pm \sigma$)	CTE $\Delta\%$
module-1	10/10	+48.0% $\pm$ 2.8%	+54.9
module-2	10/10	+48.6% $\pm$ 1.7%	+55.1
module-2.v2	10/10	+48.3% $\pm$ 1.6%	+56.8
module-3	9/10	+52.3% $\pm$ 3.3%	+61.3

• 3. Per-platform breakdown

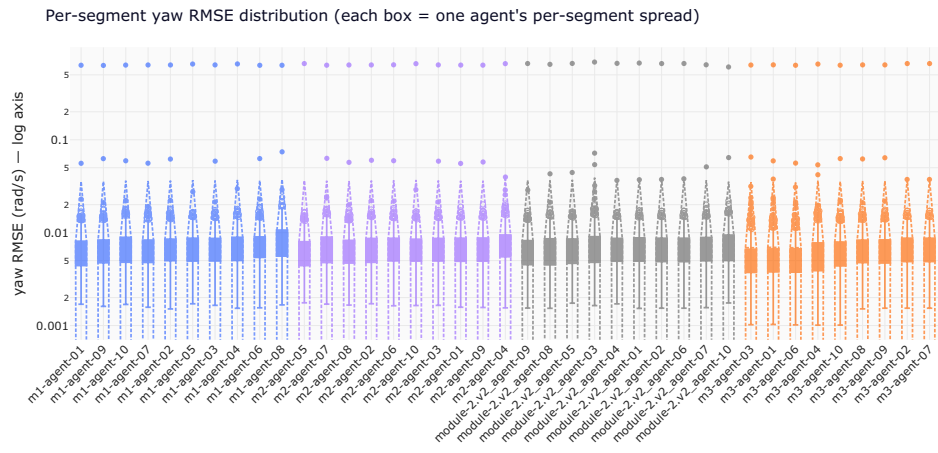
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw $\Delta\%$ (mean $\pm \sigma$)
FORD_F_150_LIGHTNING_MK1	39	+20.4% $\pm$ 2.3%
FORD_MUSTANG_MACH_E_MK1	39	+53.2% $\pm$ 5.8%
HYUNDAI_IONIQ_5	39	+53.1% $\pm$ 3.0%

• 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status
m1-agent-01	module-1	ok
m1-agent-02	module-1	ok
m1-agent-03	module-1	ok
m1-agent-04	module-1	ok
m1-agent-05	module-1	ok

agent	family	status
m1-agent-06	module-1	ok
m1-agent-07	module-1	ok
m1-agent-08	module-1	ok
m1-agent-09	module-1	ok
m1-agent-10	module-1	ok
m2-agent-01	module-2	ok
m2-agent-02	module-2	ok
m2-agent-03	module-2	ok
m2-agent-04	module-2	ok
m2-agent-05	module-2	ok
m2-agent-06	module-2	ok
m2-agent-07	module-2	ok
m2-agent-08	module-2	ok
m2-agent-09	module-2	ok
m2-agent-10	module-2	ok
module-2.v2_agent-01	module-2.v2	ok
module-2.v2_agent-02	module-2.v2	ok

agent	family	status
module-2.v2_agent-03	module-2.v2	ok
module-2.v2_agent-04	module-2.v2	ok
module-2.v2_agent-05	module-2.v2	ok
module-2.v2_agent-06	module-2.v2	ok
module-2.v2_agent-07	module-2.v2	ok
module-2.v2_agent-08	module-2.v2	ok
module-2.v2_agent-09	module-2.v2	ok
module-2.v2_agent-10	module-2.v2	ok
m3-agent-01	module-3	ok
m3-agent-02	module-3	ok
m3-agent-03	module-3	ok
m3-agent-04	module-3	ok
m3-agent-05	module-3	import_failed
m3-agent-06	module-3	ok
m3-agent-07	module-3	ok
m3-agent-08	module-3	ok
m3-agent-09	module-3	ok

agent	family	status
m3-agent-10	module-3	ok

- **6. Calibration cards — fitted coefficients**

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
```

```
    "bias_rad": 0.0
  }
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
    "b": 0.9120045284217976,
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "_default": {
    "a": 0.0,
    "b": 1.0,
    "c": 0.0
  }
}
```

m1-agent-06

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    },
    "m3": {
      "a": 0.9751496691433119,
      "b": 0.23839497273290036,
      "c": 0.0004082870644824788
    },
    "val_rmse": {
      "V0": 0.0267999381028469,
      "M1": 0.027439581036525294,
      "M2": 0.026420879339395156,
      "M3": 0.031903168689021696
    },
    "train_rmse": {
      "V0": 0.012685628695945105,
```

```

    "M1": 0.011389013310887795,
    "M2": 0.008745921115146504,
    "M3": 0.010386863485176999
  }
},
"FORD_F_150_LIGHTNING_MK1": {
  "L": 3.7,
  "best": "M2_understeer",
  "m1": {
    "a": 0.8569324797501493
  },
  "m2": {
    "L_eff": 3.882198880475702,
    "K_us": 0.003439279945738158
  },
  "m3": {
    "a": 0.8772240685582882,
    "b": -0.031353830652592465,
    "c": -0.0036354381262862774
  },
  "val_rmse": {
    "V0": 0.01591104278075153,
    "M1": 0.011141896721881754,
    "M2": 0.0073696198273352906,
    "M3": 0.0104650790814036
  },
  "train_rmse": {
    "V0": 0.02018452928972004,

```

m1-agent-08

```

{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,

```

```
    "lag": 3,  
    "rmse": 0.00888482422851739  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "K": 0.002625595512351926,  
    "s": 1.1771921499482927,  
    "off": 3.9923690213217876e-05,  
    "lag": 4,  
    "rmse": 0.01315660341959157  
  },  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L": 3.7,  
    "K": 0.0035388455508032707,  
    "s": 0.9569361755429138,  
    "off": -0.0011908292677060655,  
    "lag": 3,  
    "rmse": 0.011832340064259452  
  },  
  "TESLA_MODEL_3": {  
    "L": 2.875,  
    "K": 0.0033895612010179177,  
    "s": 1.0,  
    "off": 0.0,  
    "lag": 3,  
    "rmse": null  
  }  
}
```

m1-agent-10

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred.",
    "passthrough": true
  }
}
```

m2-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
    "n_samples": 430990,
    "rmse_v0": 0.016327157090019005,
    "rmse_v1": 0.007867193883977044,
    "K_v1": 0.0043769531249999985,
    "rmse_v2": 0.00640773025816659,
    "K_v2": 0.003836687017912508,
    "a_v2": 0.9743406550510199,
    "b_v2": -0.001228592917784131
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.984,
    "n_samples": 615175,
    "rmse_v0": 0.013616820108542612,
    "rmse_v1": 0.013930807470187448,
    "K_v1": 0.000785400390624989,
    "rmse_v2": 0.0095149036095479,
    "K_v2": 0.0028742800060782272,
    "a_v2": 1.1998919734409932,
    "b_v2": 3.385609342398125e-05
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
    "L": 3.0,
    "n_samples": 2042270,
    "rmse_v0": 0.017084454875116246,
    "rmse_v1": 0.009100616867655459,
    "K_v1": 0.0042265624999999985,
    "rmse_v2": 0.008741269113389604,
```

```
"K_v2": 0.0033859035164445724,  
"a_v2": 0.9707125684779052,  
"b_v2": 0.0005066770839987745  
}  
}
```

m2-agent-07

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "tau": 0.05,  
    "L_eff": 3.8588781722695416,  
    "K_u": 0.00314039574481963,  
    "delta_bias_rad": -0.001157331673169034,  
    "rmse": 0.005592721659455879  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "tau": 0.08,  
    "L_eff": 2.5486304264900417,  
    "K_u": 0.0015312706121448161,  
    "delta_bias_rad": 4.8217638587095315e-05,  
    "rmse": 0.008347407485517062  
  },  
  "HYUNDAI_IONIQ_5": {  
    "tau": 0.05,  
    "L_eff": 3.163750474852756,  
    "K_u": 0.0021607848879418314,  
    "delta_bias_rad": 0.0005617097326308059,  
    "rmse": 0.007713638558960502  
  }  
}
```

m3-agent-01

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06,
    "delta0": 0.00133,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.93817,
```

```

    "L_eff": 2.8871,
    "K_us": 0.0028925,
    "tau": 0.061895,
    "delta0": 0.0,
    "use_per_segment_delta0": true
  },
  "_meta": {
    "delta0_detector": {
      "method": "yaw_rate_pred_rads_low_speed_high",
      "yr_thresh": 0.03,
      "v_thresh": 5.0,
      "min_rows": 50
    },
    "notes": "Lightning coeffs hand-set from anti-patterns.md prior"
  }
}

```

m3-agent-03

```

{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.031539710821034,
    "L_eff": 2.605202152510245,
    "K_us": 0.001860730244989818,
    "tau": 0.0627605678289049,
    "delta0_fallback": 0.00016924506921824425,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8860740528359514,
    "L_eff": 3.380345065967063,
    "K_us": 0.0032867081270640908,
    "tau": 0.05844688797895399,
    "delta0": 0.001336241196109372,
    "use_per_segment_delta0": false
  }
}

```

```
},
"HYUNDAI_IONIQ_5": {
  "g": 0.896091628794913,
  "L_eff": 2.8008769715764292,
  "K_us": 0.0023750439062653513,
  "tau": 0.04533958799360384,
  "delta0_fallback": -0.00028615565156215293,
  "v_thresh": 8.0,
  "delta_thresh": 0.005,
  "min_rows": 80,
  "use_per_segment_delta0": true
}
}
```

m3-agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.0010739940929899243,
    "g": 0.8428393582368884,
    "L_eff": 3.26,
    "K_us": 0.0024711101464802783,
    "tau": 0.05493261284467375
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -9.587089484890604e-05,
    "delta0": -9.587089484890604e-05,
    "g": 0.8814740811859291,
    "L_eff": 2.22,
    "K_us": 0.001600317906539393,
    "tau": 0.08022498681872611
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 7.704255037895423e-05,
    "delta0": 7.704255037895423e-05,
```

```
    "g": 0.9863062253350734,  
    "L_eff": 3.033023238719065,  
    "K_us": 0.0028341308076772504,  
    "tau": 0.055317906584940124  
  },  
  "TESLA_MODEL_3": {  
    "passthrough": true  
  }  
}
```

m3-agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 0.6006343437133439,  
    "delta0": 0.0012709657890603524,  
    "K_us": 0.002411110430682984,  
    "tau": 0.05905789911423279,  
    "L_eff": 2.273362244343053,  
    "loss": 0.005658956187762647,  
    "n_segments_fit": 175  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "g": 0.9449514060826516,  
    "delta0": -2.802364842758471e-05,  
    "K_us": 0.0023030236477064245,  
    "tau": 0.06905315476018789,  
    "L_eff": 2.3418351435366787,  
    "loss": 0.008957550273019014,  
    "n_segments_fit": 240  
  },  
}
```

```
"HYUNDAI_IONIQ_5": {
  "g": 0.8635558962089673,
  "delta0": -0.0002751563706215791,
  "K_us": 0.0031040456441715804,
  "tau": 0.051877028201724164,
  "L_eff": 2.663564560241335,
  "loss": 0.009441409401572077,
  "n_segments_fit": 400
}
}
```

m3-agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 1.2677013760203653,
    "L_eff": 4.8869672279395875,
    "K_us": 0.004638743872488176,
    "tau": 0.056604854781974856,
    "delta0_glob": 0.0012307692372779105,
    "use_per_segment_delta0": false,
    "train_yaw_rmse": 0.012681866822813753
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.4999996985088524,
    "L_eff": 3.7976856683656175,
    "K_us": 0.0030911056729474493,
    "tau": 0.06768316418910694,
    "delta0_glob": 0.0003094591239113909,
    "use_per_segment_delta0": true,
    "train_yaw_rmse": 0.013524463171650382
  },
  "HYUNDAI_IONIQ_5": {
    "g": 1.4070879006714407,
    "L_eff": 4.3500385986664405,
    "K_us": 0.004915645105647427,
    "tau": 0.05040893663177004,
    "delta0_glob": -0.0005219437645709744,
```

```
"use_per_segment_delta0": false,  
  "train_yaw_rmse": 0.009412230675873393  
}  
}
```

module-2.v2_agent-02

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "k": 0.9366323579288449,  
    "K_us": 0.0008734759361408742,  
    "b": -0.004441819259140514  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "k": 1.1760967499879564,  
    "K_us": 0.0008734571660548825,  
    "b": 0.0001733616708132899  
  },  
  "HYUNDAI_IONIQ_5": {  
    "k": 0.9342137556967759,  
    "K_us": 0.0009874770996559546,  
    "b": 0.002002086108887664  
  },  
  "TESLA_MODEL_3": {  
    "k": 1.0,  
    "K_us": 0.0,  
    "b": 0.0  
  }  
}
```

module-2.v2_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "d0": 0.0012435085448964291,
    "kk": 0.9364948189665847,
    "KK": 0.0032470115980856545,
    "tau": 0.0
  },
  "HYUNDAI_IONIQ_5": {
    "L": 2.97,
    "d0": -0.0005236657821362199,
    "kk": 0.933574693892375,
    "KK": 0.0029030763922389792,
    "tau": 0.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "d0": -2.675943355525215e-05,
    "kk": 1.1761809849040472,
    "KK": 0.0026070689183535123,
    "tau": 0.0
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "d0": 0.0,
    "kk": 1.0,
    "KK": 0.0,
    "tau": 0.0
  }
}
```

module-2.v2_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9397533271985806,
    "K": 0.0008518149458872777,
    "b": -0.0044345248756529525
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 1.1808972441892185,
    "K": 0.0008794604321610935,
    "b": 0.00016328778675285131
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.9350770580282316,
    "K": 0.0009883221917463027,
    "b": 0.002004778269326024
  },
  "TESLA_MODEL_3": {
    "a": 1.0,
    "K": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "G": 0.9041237868107276,
    "Kus": 0.0007955703708621087,
    "bias": -0.005088051550305564,
    "tau": 0.05804217686893913,
    "kdot": -0.012302023150477267
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "G": 1.1560312029550013,
    "Kus": 0.0010408901778361697,
    "bias": 0.0007331723308643077,
    "tau": 0.040688651274880756,
    "kdot": 0.008800778388725197
  },
  "HYUNDAI_IONIQ_5": {
    "G": 0.9700159639975359,
    "Kus": 0.001582087902935487,
    "bias": 0.00211745029940103,
    "tau": 0.04998898336470954,
    "kdot": 3.516097849258846e-05
  },
  "TESLA_MODEL_3": {
    "G": 1.0,
    "Kus": 0.0,
    "bias": 0.0,
    "tau": 0.0,
    "kdot": 0.0
  }
}
```

module-2.v2_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.0009299897686168917,
```

```
    "g": -0.13146377934999154,  
    "b": -0.004858365647164137  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "k": 0.0007776286604362939,  
    "g": 0.1760123050495998,  
    "b": 0.0007151284926824161  
  },  
  "HYUNDAI_IONIQ_5": {  
    "k": 0.0017086888155224455,  
    "g": -0.02937372993329363,  
    "b": 0.0030989091566628654  
  },  
  "TESLA_MODEL_3": {  
    "k": 0.0,  
    "g": 0.0,  
    "b": 0.0  
  }  
}
```

● 6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

Clean cohort: no agent referenced truth-derived column names inside their predict() body.

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	40	0
has_manifest_json	40	0
manifest_parsable	40	0
manifest_declares_predict_callable	40	0
manifest_declares_platform_support	40	0
has_predict_py	40	0
has_coeffs_json	33	7

format check	pass	fail
has_report	8	32

Failure reasons across the cohort:

- import_failed — 1 agent(s)

• 8. Worst-of-cohort

Lowest yaw $\Delta\%$

- m1-agent-08
(+40.4%)
- m2-agent-04
(+44.6%)
- module-2.v2_agent-03
(+45.3%)

Lowest CTE $\Delta\%$

- m1-agent-10
(+49.5%)
- m2-agent-07
(+49.5%)
- m2-agent-04
(+50.0%)

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